Quality Management

7 — Analog image formation: light, lenses and the basics of photography

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Today

How an image forms — before it becomes numbers

Every image used for inspection starts the same way: **light** leaves the part, a **lens** collects it, and an image forms on a surface. Today: that analog half of the chain, and the camera settings that decide what the image shows. Next lecture: how the image becomes numbers.

1. **Light and the eye**
   spectrum, colour, the first imaging system
2. **Forming an image**
   pinhole, lens, focusing
3. **Field and focus**
   focal length, perspective, aperture, depth of field
4. **Exposure**
   time, light, gain — and what each costs
5. **Lens errors**
   distortion, perspective, telecentric lenses

Why images in quality management

Looking is the oldest inspection there is

Long before gauges and sensors, quality was checked by **looking**: a weaver at the cloth, a smith at the blade, an inspector at the end of the line. Visual inspection is still in every quality plan — and most of it is now done by cameras.

What is checked by looking	Example
dimensions and shape	diameters, positions of holes, gaps
surface	scratches, dents, porosity, paint runs
completeness and assembly	missing clip, seal the wrong way round
colour and appearance	paint shade, ripeness of fruit
marking	labels, dates, codes

The same physics for the eye and the camera

Both are imaging systems: a lens, an aperture, a light-sensitive surface. Understanding how an image forms tells you what a camera — or an inspector — can and cannot see, and how to set it up so that the defect is in the image.

What changes with the camera

It does not get tired, it checks every part, and it gives the same answer twice — but only if light, distance, focus and exposure are controlled. Those controls are today's subject.

Two centuries of images that measure

From the camera obscura to the camera on the line

When	What	Why it mattered
antiquity – 1020	camera obscura; Alhazen's *Book of Optics*	light through a hole forms an image — the pinhole model
1826	first permanent photograph, Niépce (bitumen on a plate, hours of exposure)	an image can be recorded
1839	daguerreotype: silver-plated copper, minutes of exposure	the first practical photography
1850s–1860s	Laussedat and Meydenbauer: **photogrammetry**, measuring buildings from photographs	**images used to measure**
1888 · 1907	Kodak roll film · Autochrome, the first practical colour process	photography for everyone, in colour
1969 · 1975	CCD sensor, Boyle and Smith (Nobel Prize 2009) · first digital camera, Sasson at Kodak: 0.01 megapixels	the image becomes numbers (next lecture)
1980s	machine vision and industrial close-range photogrammetry	cameras on production lines
1990s – today	CMOS image sensors; phones; computational imaging	cheap, fast cameras everywhere

Through all of it, the optics did not change: the pinhole and the lens of 1826 obey the same laws as the camera on today's line.

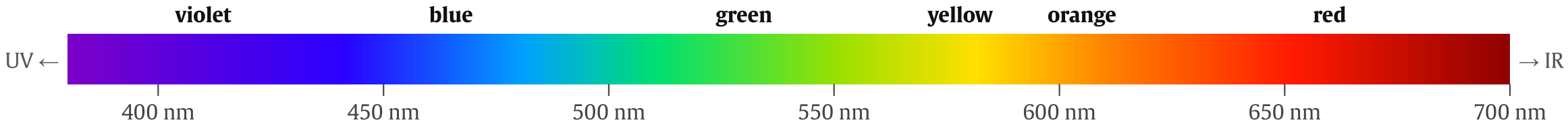
PART 1

Light and the eye

The visible spectrum · The eye as a camera · Colour vision · What the eye gets wrong

Light and colour

What an image is made of



white light: all of them together

Light is a wave

Visible light is the narrow band of electromagnetic waves from about **380 to 700 nm**. Each wavelength is seen as a colour; white light is a mixture of all of them.

An object's colour

A surface returns part of the light it receives and absorbs the rest. A red part returns red and absorbs the others: its colour depends on the part **and** on the light.

Beyond the visible

Ultraviolet below 380 nm, infrared above 700 nm: invisible to the eye, but not to cameras — useful in inspection (lectures 9 and 10).

The eye: the first imaging system

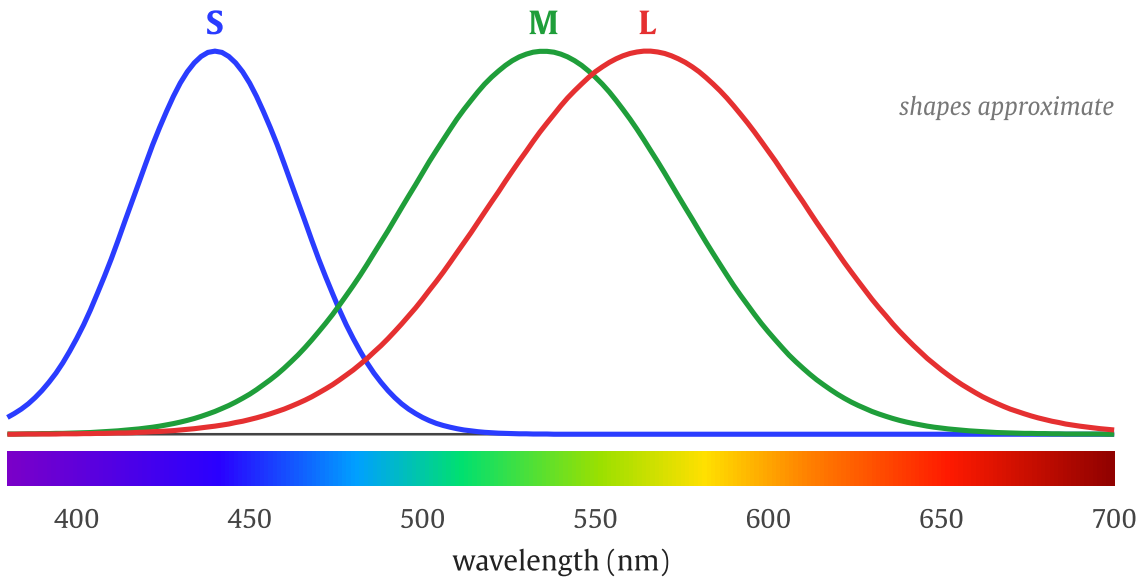
Every part of a camera has its counterpart in the eye

Eye	Camera	Job
cornea and crystalline lens	lens	forms the image
iris and pupil (2–8 mm)	aperture	lets in more or less light
changing the lens's shape	moving the lens	focusing
retina	film or sensor	records the image
rods and cones	pixels	sense the light, point by point
brain	image processing	interprets, corrects, decides

- The eye's lens has a focal length of about 17 mm. With the pupil between 2 and 8 mm, the eye works between about **f/8 and f/2** — the same range as a camera lens.
- About 120 million **rods** sense brightness, even in dim light. About 6 million **cones**, packed in the centre, sense colour and fine detail.
- The eye focuses by changing the shape of its lens; a camera does it by moving the lens. Both obey the same lens law (slide 14).

How we see colour

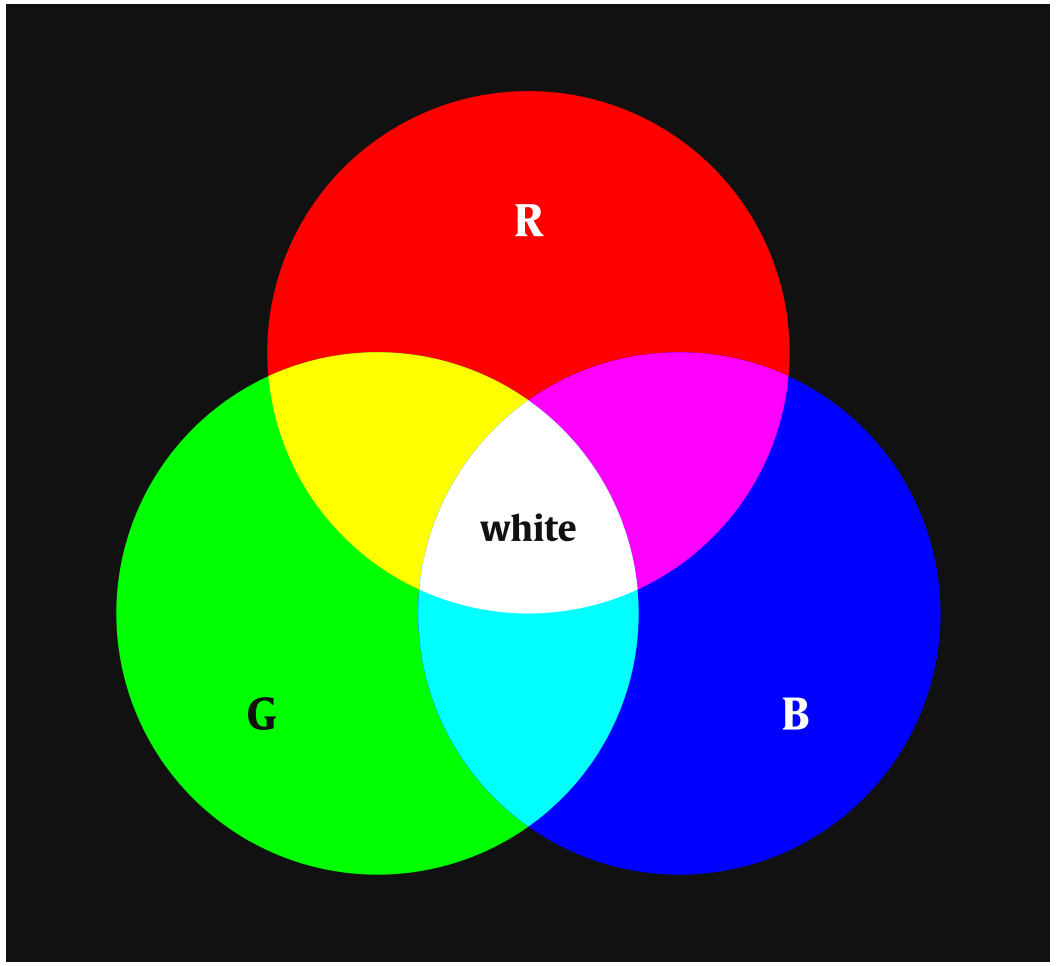
Three kinds of cone, and colour from their ratios



- Three types of cone, most sensitive to **short** (blue), **medium** (green) and **long** (red) wavelengths. Their sensitivities overlap.
- A single cone cannot tell colour: it only responds more or less. Colour is the brain's reading of the **ratio** between the three responses — *trichromatic* vision.
- So any colour we see can be matched by mixing **three** lights. This is why screens, cameras and colour images use three channels — red, green and blue. Next lecture builds on it.

Mixing colours: light and ink

Why screens use RGB and printers use CMYK



Additive: adding light
Red, green and blue lights add up: red + green = yellow, all three = white. Screens, projectors and cameras work this way.

Subtractive: removing light
Each ink absorbs part of the white light falling on it. Cyan, magenta and yellow inks remove red, green and blue; all three give (almost) black, so printers add a black ink: **CMYK**.

In quality control both matter: a camera judges colour additively, a printed label or a painted part makes its colour subtractively — and both depend on the light they are seen under.

What the eye gets wrong

Why visual inspection by people needs rules — and why cameras took over

Effect	What happens
Colour constancy	the brain corrects for the light: white paper looks white under a lamp and in daylight
Simultaneous contrast	the same grey looks lighter on a dark background than on a light one
Metamerism	two paints match under one light and differ under another
Colour deficiency	about 8 % of men and 0.5 % of women see some colours differently
Fatigue and attention	defects are missed more often as the shift goes on, and rare defects most of all
Judgement	two inspectors, or one inspector on two days, do not always agree

When people inspect

Standard daylight in a light booth, with neutral grey surroundings (ISO 3664) · **limit samples** of acceptable and rejected parts · colour-vision tests · breaks and rotation.

What a camera fixes — and what it does not

It does not adapt, tire or disagree with itself. But it has no colour constancy either: it records exactly the light it gets, so the light must be controlled (lecture 10).

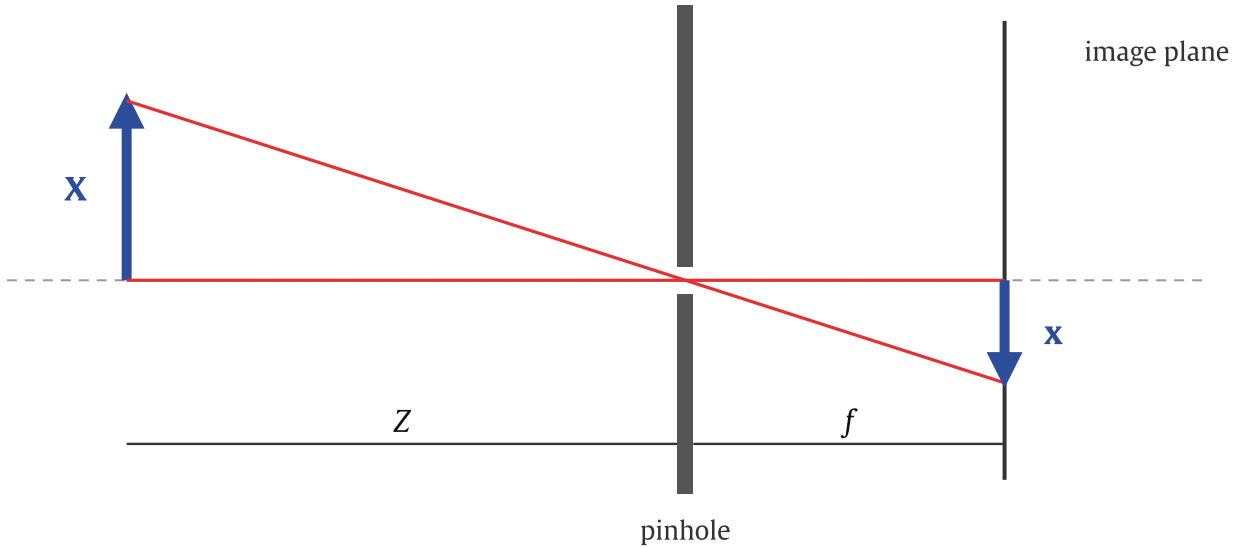
PART 2

Forming an image

Pinhole · The pinhole's dilemma · Lens · Focusing

The pinhole camera

The simplest model of image formation — and still the one used to calibrate cameras



$x = f \cdot X / Z$

magnification $m = x / X = f / Z$

- A dark box with a small hole: each point of the object sends light through the hole to **one** point of the back wall. The image is sharp and **inverted**.
- Its size follows from **similar triangles**: an object of height X at distance Z gives an image $x = f\,X/Z$, with f the distance from hole to image.
- The magnification f/Z depends on the distance: **the same part looks bigger when it is closer**. For measurement, that is an error to control (slide 30).

The pinhole's dilemma

A big hole blurs, a tiny hole diffracts — and neither lets in much light



2 mm | **0.8 mm** | **0.33 mm — best** | **0.1 mm** | **0.05 mm**

← *hole too large: blur* — *hole too small: diffraction* →

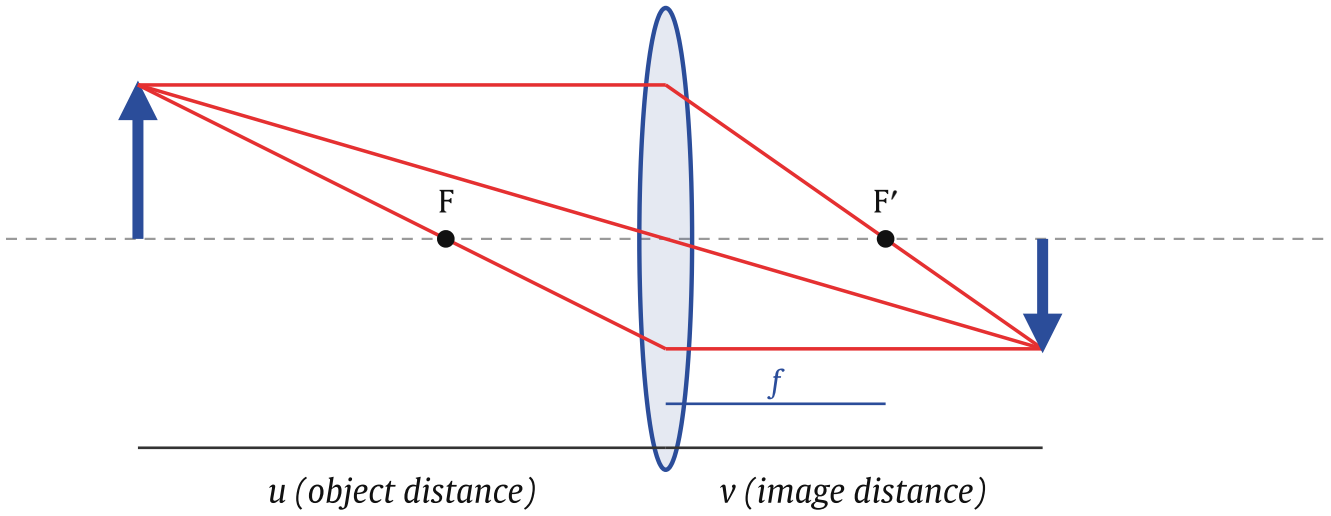
- **Large hole:** every point of the object becomes a disc as large as the hole — the image blurs.
- **Small hole:** light spreads out as it passes a narrow opening — **diffraction** — and the image blurs again.
- The sharpest image comes from a hole of about $d \approx 2\sqrt{(f\lambda)}$: 0.33 mm for $f = 50$ mm in green light.

Why we need a lens

Even the best pinhole lets in so little light that an exposure lasts seconds or minutes. A **lens** collects light over a wide opening and still brings each point to one point — sharp *and* bright.

The thin lens

Where the image forms, and how large it is



$$1/f = 1/u + 1/v \qquad m = v/u$$

- Three rays find the image: parallel to the axis, then through F′; through the centre, undeviated; through F, then parallel.
- **Converging** (positive) lenses form real images — the ones in cameras and in the eye. **Diverging** (negative) ones spread light: glasses for short sight, and correcting elements inside camera lenses.
- A camera lens has many glass elements, to correct its errors (slide 29), but for choosing it you treat it as one thin lens of focal length f.

Where f comes from

The glass's refractive index n and the two surface curvatures R_1, R_2: the lensmaker's equation, $1/f = (n-1)(1/R_1 - 1/R_2)$. Designers use it; users read f on the lens.

Focusing, from far to close up

The closer the object, the further behind the lens its image

From the lens law, $v = u\,f / (u - f)$. For a 25 mm lens:

Object at u	Image at v	Magnification
very far	25.0 mm	≈ 0
1000 mm	25.6 mm	0.026
321 mm	27.1 mm	0.084
100 mm	33.3 mm	0.33
50 mm ($= 2f$)	50 mm	**1 : 1**

- **Focusing** moves the lens away from the sensor as the object comes closer. Every lens has a **minimum object distance**: below it, it can no longer move far enough.
- For small parts, **extension rings** or **macro lenses** move the lens further out and give magnifications near 1 : 1 — the part as large on the sensor as in reality.
- In inspection the distance is fixed: focus once, **lock the focus ring**, and check it with a reference image — a lens that slowly drifts out of focus is a gauge that drifts.

Recap TAKEAWAYS

Parts 1 and 2 — light, the eye, forming an image

1. **Light** from 380 to 700 nm; an object's colour depends on the object and on the light.
2. **The eye is a camera:** lens, aperture (f/2–f/8), retina. Three cone types → three channels are enough to reproduce colour.
3. **The eye is not an instrument:** constancy, contrast effects, metamerism, colour deficiency, fatigue. People inspect under rules; cameras under controlled light.
4. **Pinhole:** similar triangles, $m = f/Z$. The hole size trades blur against diffraction, and gives little light.
5. **Lens:** $1/f = 1/u + 1/v$, $m = v/u$. Focus is set once and locked in inspection.

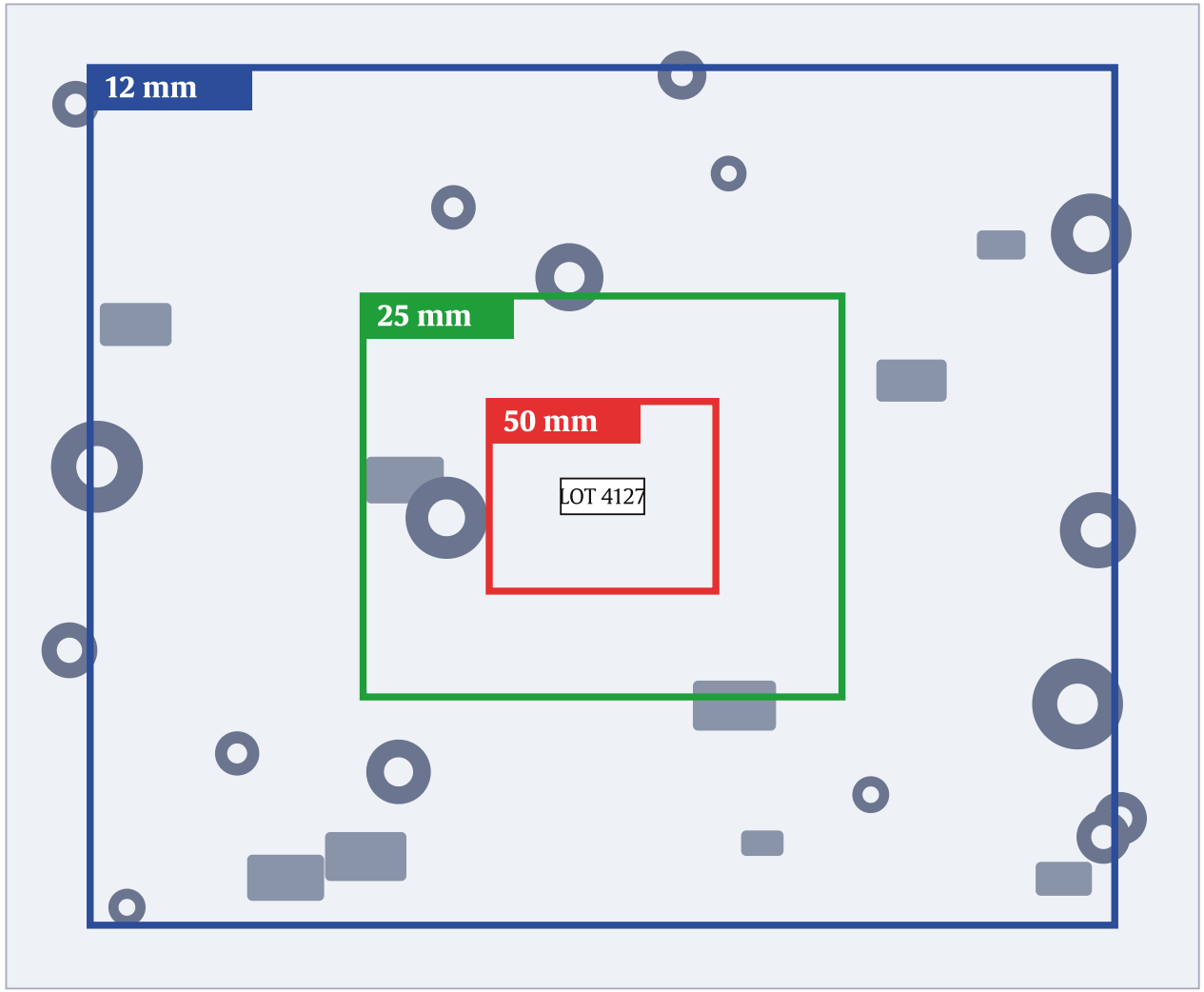
PART 3

The basics of photography

Focal length · Perspective · Aperture · Depth of field · Exposure · Gain

Focal length and field of view

Same camera, same distance, three lenses



With a sensor of width S, a lens of focal length f at distance u sees a field of width

$$W = S(u - f) / f \qquad \text{angle} = 2\arctan(S/2f)$$

Lens	Angle of view	Field at 500 mm
12 mm	39°	344 mm
25 mm	19°	161 mm
50 mm	10°	76 mm

Sensor 8.45 mm wide. Twice the focal length: half the field, twice the detail on every part of it.

To see more of the scene: shorter focal length, or step back. **To see a detail larger:** longer focal length, or come closer.

Distance, focal length and perspective

The same framing can be obtained in two ways — they do not give the same image

- A field of 100 mm can be framed with a **short lens from close** or a **long lens from far**. The field is the same; the perspective is not.
- Perspective depends only on the **distance**: a feature 10 mm nearer the camera looks larger by about $10/u$.
- From close, near and far parts of the object look very different in size, and the sides of a box become visible. From far, the object looks "flat".

Camera at	Feature 10 mm nearer looks
100 mm	11 % larger
200 mm	5 % larger
1000 mm	**1 % larger**

For measurement

Step back and use a longer lens: the scale changes less with the height of the part. Taken to the limit, this is the telecentric lens (slide 30).

Aperture and f-number

How wide the lens opens

$N = f / D$ with D the diameter of the opening. The light reaching the sensor goes as $1/N^2$: each step of the standard sequence multiplies N by $\sqrt{2}$ and **halves the light** — one **stop**.



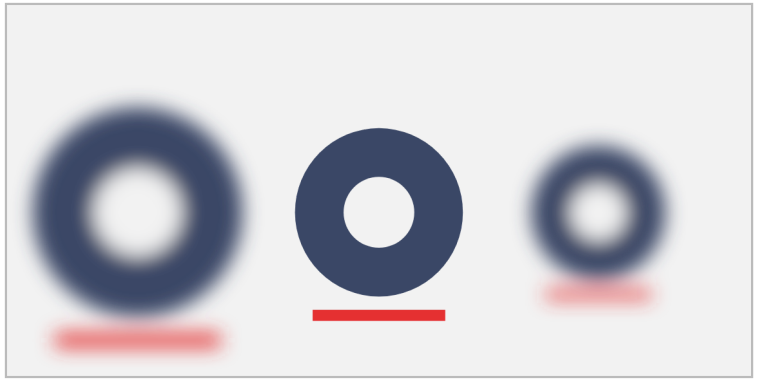
Light
Small N (wide open): more light, shorter exposure.

Depth of field
Large N: more of the scene in focus (next slide).

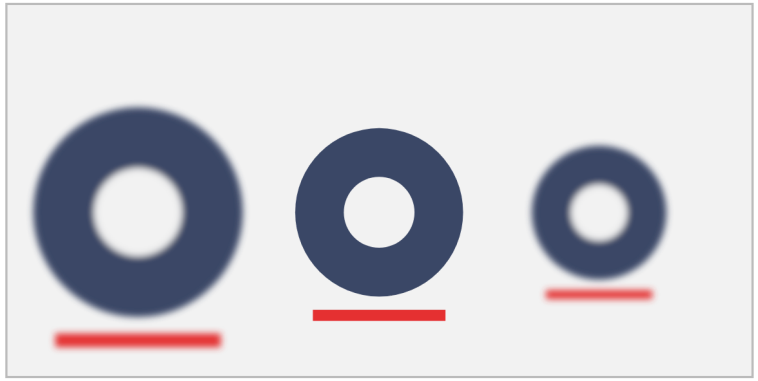
Diffraction
Large N: every point blurs into a disc of diameter $\approx 2.44\,\lambda\,N$ (slide 22).

Depth of field

How much of the scene, in front of and behind the focus, still looks sharp



f/2: only the middle part is sharp



f/5.6: more is sharp



f/16: all sharp, but all a bit soft

$$\mathrm{DoF} \approx 2\, N\, c\, (1 + m) \,/\, m^2$$

c: the largest blur accepted at the sensor (the *circle of confusion*) · m: magnification. Depth of field grows with N, and grows fast as the magnification falls — close-ups have almost none.

- **Large depth:** to keep a tall part, or parts at different heights, all sharp.
- **Small depth:** to isolate one plane — the surface of a part from the cluttered fixture behind it.
  - Photographers' *hyperfocal distance* $H \approx f^2/(N\,c)$: focused there, everything from $H/2$ to infinity is sharp.

The price of stopping down

Depth of field against diffraction: stop down as far as you need, not as far as you can

N	Light vs $N = 4$	Depth of field	Diffraction disc
2.8	× 2	5.9 mm	3.8 µm
4	× 1	**8.4 mm**	**5.4 µm**
5.6	× ½	11.7 mm	7.5 µm
8	× ¼	16.8 mm	10.7 µm
11	× ⅛	23.1 mm	14.8 µm
16	× 1/16	33.5 mm	21.5 µm

25 mm lens at 321 mm ($m = 0.0845$), accepted blur $c = 6.9$ µm, green light.

- Closing the lens buys depth of field, but every step costs half the light **and** enlarges the diffraction disc.
- When the disc grows larger than the blur you accept, the whole image gets softer — even the plane in focus. At $N = 16$ it is three times larger than c.

The rule

Find the depth you need; choose the smallest N that gives it; make up the light with the lighting, not with the gain.

Exposure time and motion

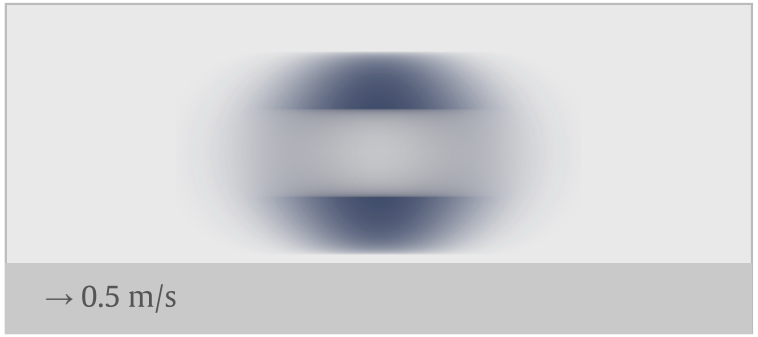
How long the shutter stays open — and what a moving part does meanwhile



1/12 000 s · strobe



1/1000 s



1/100 s

During the exposure time t a part moving at speed v travels $v \cdot t$. If that is more than the smallest detail you need, the image is blurred along the motion.

Exposure, at 0.5 m/s	Part moves
1/100 s	5 mm
1/1000 s	0.5 mm
1/12 000 s (82 µs)	**0.04 mm**

- To freeze a part that moves, shorten the exposure: but each halving costs half the light.
- On production lines the answer is a **strobe**: a flash of light far shorter than the exposure, synchronised with the camera. The flash, not the shutter, sets the effective exposure time.
- For parts that stand still, exposure time is free: use it to collect light and keep the aperture and the gain where they should be.

Exposure: aperture and time together

The same amount of light can be collected in many ways



under-exposed



well exposed



over-exposed

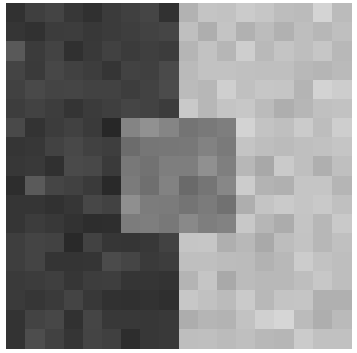
Equal exposures: one stop of aperture traded for one of time			
f/4 · 1/500 s	f/5.6 · 1/250 s	f/8 · 1/125 s	f/11 · 1/60 s
least depth, freezes motion			most depth, risks blur

Photographers write the setting as an exposure value, EV = $\log_2(N^2/t)$: f/8 at 1/125 s is EV 13.

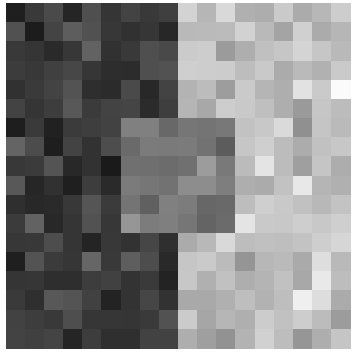
- **Under-exposed:** dark parts sink into black and into the noise.
- **Over-exposed:** bright parts reach the maximum the film or sensor can record — **saturation**. The detail in them is lost for good, and edges move.
- Set the exposure from the **brightest region you need**, just below saturation.

Gain (ISO) and noise

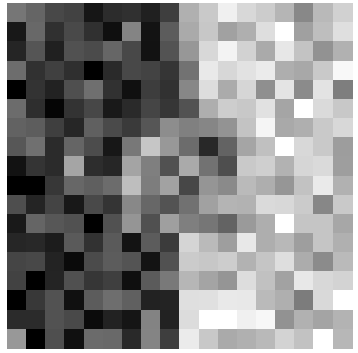
The third control — and the one that does not add light



plenty of light, gain × 1



¼ of the light, gain × 4



1/16 of the light, gain × 16

- **ISO** once meant the sensitivity of a film; in a camera it is the **gain** applied to the signal. Doubling it makes the image as bright as one stop more light.
- But it amplifies the noise together with the signal: an image made bright by gain is **noisier** than one made bright by light. Lecture 9 explains why.
- High gain also lowers the brightest value that can be recorded, so highlights saturate sooner.

Order of preference

More light → longer exposure (if nothing moves) → wider aperture (if the depth allows) → gain, last.

Setting the camera for what you need to see

Every change has a price, and something else to compensate it

You need…	Change	It costs	Compensate with
a tall part sharp from top to bottom	larger N	light; diffraction	more light; stop at the depth you need
a moving part without blur	shorter exposure	light	a strobe; a wider aperture if the depth allows
a dark image to be bright	more light first	—	then longer exposure; gain last
more of the scene in the image	shorter focal length, or step back	less detail on each part	a higher-resolution sensor (lecture 9)
a small detail larger	longer focal length, or come closer	smaller field; less depth	larger N; several images
the same scale at every height	step back with a longer lens	a longer set-up	a telecentric lens
one plane, without the clutter behind	smaller N (shallow depth)	less tolerance on position	better: a backlight or dark field (lecture 10)
no hot spots on a shiny part	shorter exposure	darker elsewhere	diffuse light, polarisers (lecture 10)

Recap TAKEAWAYS

Part 3 — the basics of photography

1. **Focal length** sets the field: $W = S(u - f)/f$. **Distance** sets the perspective: step back with a longer lens to measure.
2. **Aperture** $N = f/D$: one stop halves the light. Larger N, more depth of field — and more diffraction.
3. **Exposure time** trades light against motion blur; a strobe freezes moving parts.
4. **Exposure** is set from the brightest region you need, just below saturation. **Gain** adds brightness, not light: use it last.
5. **Every setting has a price**: decide from what you need to see, and compensate with the light.

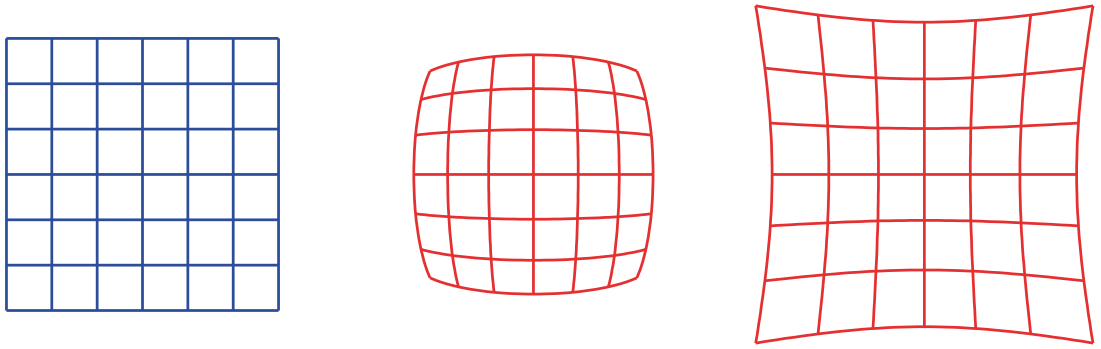
PART 4

Lens errors

Distortion · Vignetting · Chromatic aberration · Perspective · Telecentric lenses

Lens errors that change a measurement

Hard to see at a glance; all of them move an edge



none **barrel** **pincushion**

Distortion: the magnification changes with distance from the centre. Straight lines bend outwards (barrel, typical of short lenses) or inwards (pincushion). A distortion of 0.5 % moves a point 0.25 mm at the edge of a 100 mm field — more than many tolerances.

Vignetting
The image darkens towards the corners. Brightness that changes across the field moves edges found with a fixed brightness level.

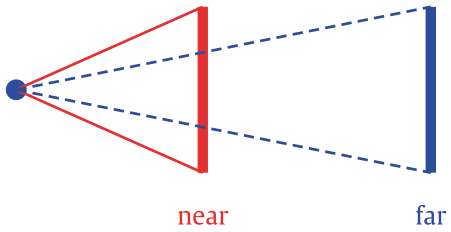
Chromatic aberration
Glass bends each wavelength differently, so colours focus at slightly different distances: coloured fringes along edges. Remedy: light of one colour.

What to do about them
Choose a low-distortion lens for gauging; then **calibrate**: image a target of known geometry, such as a dot grid, and correct the image.

Perspective, and telecentric lenses

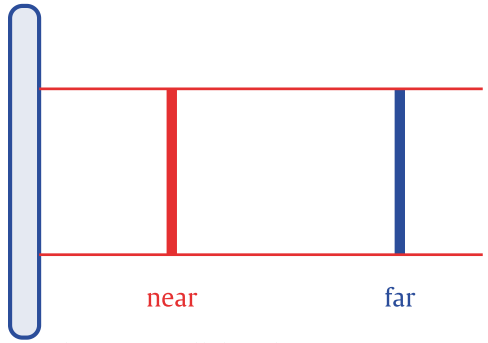
An ordinary lens sees nearer things bigger. For gauging, that is an error

ordinary lens



same part, larger angle when nearer

telecentric lens



only rays parallel to the axis: same size

- An ordinary lens accepts rays from a cone: the magnification depends on the distance (slide 19). A part 1.6 mm thick seen from 321 mm is 0.5 % larger on its top face than on its bottom one.
- A **telecentric lens** accepts only rays parallel to its axis: within its depth, the magnification does not change with distance.
- Its front lens must be as large as the field of view, so it is large and expensive — used to gauge small parts precisely.
- Without one: **backlight** the part, keep the camera **parallel** to it, and **calibrate on a master** of known size.

Putting it together: a camera to gauge washers

The optics of the hands-on lab, decided one setting at a time

M8 washers to ISO 7089: hole 8.40–8.62 mm, outside 15.57–16.00 mm, 1.6 mm thick. About twenty of them on a light box, one camera above, sensor 8.45 mm wide.

Decision	Choice	Why
Field of view	100 mm	about twenty washers per image
Lens and distance	25 mm lens at 321 mm	magnification $S/W = 0.0845$; a standard focal length
Aperture	$N = 4$	depth of field 8 mm against 1.6 mm thickness; diffraction 5.4 µm
Exposure time	whatever fills the image just below saturation	the washers stand still: no motion blur
Gain	minimum	the light box gives plenty of light
Light	backlight	a sharp silhouette, no reflections (lecture 10)
Perspective and distortion	camera parallel to the table; calibration on a master washer and a dot grid	the top face is 0.5 % larger; the lens bends the field

Still open: how many pixels, how many grey levels, which file — the digital half, next lecture.

Recap TAKEAWAYS

Part 4 — lens errors

1. **Distortion** changes the scale across the field; **vignetting** its brightness; **chromatic aberration** puts colour fringes on edges.
2. **Perspective** changes the scale with distance. Step back with a longer lens, or use a telecentric lens.
3. **Calibrate** on targets of known geometry, and keep the set-up fixed once calibrated: distance, focus, aperture.
4. **The analog image is decided before any pixel exists**: what the lens, the aperture and the exposure leave out cannot be recovered later.

Next
Digital images: how the image on the sensor becomes a matrix of numbers.

The hands-on lab
You set up these optics with your phone and a light box.

Questions?

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github.com/LolloCappo/student-hub